Beyond Emissions:
How Rainfall and Atmospheric Conditions Shape Delhi's Air Quality (2018–2024)

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ES-2302 Delhi's Air pollution

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1. Introduction

Delhi's air pollution crisis has become one of the defining environmental problems of modern urban India. Every winter, headlines describe toxic smog, hazardous Air Quality Index (AQI) levels, school closures, and public health emergencies. Public discussions surrounding this crisis usually focus on emissions alone. Vehicles, industrial activity, construction dust, coal combustion, and agricultural burning are commonly presented as the primary explanations for Delhi's severe pollution.

These sources are undeniably important. However, this dominant narrative can unintentionally simplify how pollution actually behaves in the atmosphere. Pollution is not only determined by how much is emitted. Atmospheric conditions strongly influence whether pollutants disperse, accumulate near the ground, or are removed from the air entirely.

This study investigates Delhi's air quality variability between 2018 and 2024 using two independent datasets: Central Pollution Control Board (CPCB) daily AQI observations and US Embassy $PM_{2.5}$ concentration measurements ($\mu g/m^3$). Together, these datasets provide complementary perspectives on the same atmospheric system. ERA5 meteorological reanalysis data is used throughout to examine the role of rainfall and wind.

The central argument of this paper is not that emissions are unimportant. Instead, the analysis demonstrates that emissions alone cannot explain the dramatic seasonal and day-to-day variability observed in Delhi's air quality. Atmospheric conditions actively regulate whether pollution accumulates or is removed.

This means Delhi's pollution crisis should not be understood only as an emissions crisis. It is also an *atmospheric accumulation crisis*.

The COVID-19 lockdown period provides a particularly useful natural experiment because human activity decreased sharply while atmospheric processes continued operating normally. This allows the study to partially separate the role of emissions from the role of meteorology.

2. Research Question and Hypotheses

This project investigates the following question:

> *How strongly do rainfall and atmospheric conditions influence Delhi's air quality between 2018 and 2024, and how does this complicate dominant narratives that explain pollution primarily through emissions alone?*

2.1 Null and Alternate Hypotheses

Two formal hypotheses structure the quantitative analysis:

Null Hypothesis (H_0): Rainfall, wind, and seasonal atmospheric conditions have no statistically significant relationship with Delhi's air quality (AQI or $PM_{2.5}$ concentrations). Observed variability in air quality is explained entirely by variation in anthropogenic emissions.

Alternate Hypothesis (H_1): Rainfall, wind, and seasonal atmospheric conditions have a statistically significant negative relationship with Delhi's air quality. Meteorological processes independently explain a significant portion of observed air quality variability, beyond what can be attributed to emissions alone.

These hypotheses are tested using Spearman rank correlations, one-way ANOVA, independent t-tests, and lagged correlation analysis across both the CPCB AQI and US Embassy $PM_{2.5}$ datasets.

3. Background

Most public discussions surrounding Delhi's air pollution focus on identifying pollution sources. Government interventions often emphasize reducing traffic emissions, regulating industrial output, controlling construction dust, or limiting agricultural burning. These measures are important because emissions provide the material basis for pollution formation.

However, atmospheric science demonstrates that emissions alone do not directly determine observed air quality. Once pollutants enter the atmosphere, meteorological conditions regulate whether they disperse, remain trapped near the surface, or are removed through atmospheric cleansing processes [4, 2].

Rainfall removes airborne pollutants through wet deposition, in which falling raindrops physically scavenge particulate matter and soluble gases from the atmosphere. Seasonal circulation patterns also strongly influence pollution accumulation. During winter, stable atmospheric conditions, weak winds, and reduced vertical mixing allow pollutants to remain concentrated near the surface. During monsoon conditions, stronger circulation and rainfall promote atmospheric cleansing.

This distinction is important because Delhi's emissions do not decrease dramatically during monsoon months, yet AQI values and $PM_{2.5}$ concentrations decline substantially. Such large seasonal differences suggest that atmospheric conditions strongly amplify or suppress pollution severity.

The COVID-19 lockdown period provides an additional opportunity to study this relationship. During lockdown conditions, transportation and industrial activity decreased sharply across India [3]. This created a rare large-scale reduction in anthropogenic emissions while atmospheric processes continued operating normally. The lockdown therefore functions as a natural experiment for understanding how air quality behaves when emissions decrease but meteorological processes remain active.

4. Data and Methods

4.1 AQI Data (CPCB)

Daily air-quality data for Delhi were obtained from the Central Pollution Control Board (CPCB) between January 2018 and June 2024 [1]. The original dataset consisted of monthly Excel files containing hourly AQI observations for each day. These files were processed in Python using Pandas and NumPy. Hourly AQI values were averaged to produce daily mean AQI values. The final dataset contained 2,247 daily observations spanning more than six years. A 92-day data gap exists from 1 October to 31 December 2022 due to missing source files; days within this gap are excluded from all analyses.

4.2 Why AQI Was Used Instead of Individual Pollutant Concentrations

This study focuses on the Air Quality Index (AQI) as the primary CPCB variable for both practical and conceptual reasons. According to the CPCB, AQI is a composite index designed to communicate overall air-quality severity using multiple pollutants simultaneously [1]. The Indian AQI framework combines $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, NH_3, O_3, and Pb into a single standardised scale. AQI therefore represents the integrated atmospheric pollution burden experienced by residents rather than a single pollutant concentration.

Meteorological processes such as rainfall and atmospheric stagnation influence not only particulate matter but broader atmospheric pollution conditions simultaneously. Using AQI allows the study to examine the interaction between meteorology and integrated urban air quality. Importantly, AQI is not a direct physical concentration measurement; this study interprets changes in AQI primarily as changes in overall atmospheric pollution conditions.

4.3 US Embassy $PM_{2.5}$ Data

To validate and complement the CPCB AQI data, daily $PM_{2.5}$ concentration measurements (μg/m^3) from the US Embassy monitoring station in New Delhi were obtained for the full study period [5]. The Embassy station is located in Chanakyapuri and uses a MetOne BAM-1020 Federal Equivalent Method monitor, processed under US EPA quality control protocols. The dataset contains 2,265 daily observations across the study period (1 January 2018 to 2 June 2024), providing 96.6% coverage. Unlike CPCB AQI, the Embassy data reports raw $PM_{2.5}$ in μg/m^3, confirmed by the presence of values exceeding 500 μg/m^3 (which would be impossible on any AQI scale).

4.4 Cross-Dataset Validation

The two datasets overlap on 2,168 shared days. Pearson and Spearman correlations were computed between daily CPCB AQI and Embassy $PM_{2.5}$ values on these shared days to assess whether both datasets capture the same underlying atmospheric signal. The exceptional correlation between them (Pearson $r = 0.925$; Spearman $r = 0.943$, both $p < 0.001$) demonstrates that both measure the same pollution cycle, despite different instruments, locations, and measurement approaches. This cross-validation strengthens confidence in findings drawn from either dataset independently.

4.5 Meteorological Data

Meteorological variables were obtained from the ERA5 atmospheric reanalysis dataset developed by the European Centre for Medium-Range Weather Forecasts (ECMWF) [2]. The analysis focused primarily on daily rainfall totals and daily wind component values. ERA5 hourly values were aggregated into daily quantities and merged with both air quality datasets using matching dates.

4.6 Seasonal Classification

To better represent Delhi's atmospheric structure, the year was divided into four physically meaningful seasons (Table 1). This classification reflects major seasonal shifts in rainfall, atmospheric circulation, and pollution accumulation across northern India.

Table 1: Seasonal Classification

Season	**Months**
Winter	November – February
Pre-Monsoon	March – June
Monsoon	July – September
Post-Monsoon	October

4.7 Statistical Analysis

Several complementary statistical analyses were performed. First, seasonal and monthly climatologies were calculated for both datasets. Second, Spearman rank correlation analysis was used to investigate relationships between air quality and meteorological variables. Spearman correlation was selected over Pearson because AQI and $PM_{2.5}$ data are right-skewed and rainfall data are zero-inflated (approximately 54% of days had zero recorded rainfall), violating the normality assumptions required by Pearson. Third, lagged rainfall correlations were computed to determine whether rainfall effects persist beyond individual storm events. Fourth, independent t-tests were used to compare air quality during COVID lockdown and non-lockdown conditions. Fifth, extreme-event analysis compared meteorological conditions during the cleanest and most polluted 5% of days. All statistical analysis was performed in Python using Pandas, NumPy, SciPy, and Matplotlib.

5. Results

5.1 Overall Variability

The CPCB dataset spans from 7 January 2018 through 2 June 2024 and contains 2,247 daily AQI observations. The Embassy $PM_{2.5}$ dataset covers the same study window with 2,265 daily observations. Summary statistics for both datasets are presented in Table 2.

Table 2: Overall Air Quality Statistics, 2018–2024

Statistic	**CPCB AQI**	**Embassy $PM_{2.5}$ (μg/m^3)**
Mean	204.84	162.94
Standard Deviation	102.14	74.83
Minimum	43.54	7.00
Maximum	474.92	553.00
Median	189.46	150.00
Observations	2,247	2,265

Both datasets demonstrate extremely large variability over time. The CPCB AQI standard deviation of 102.14 and Embassy $PM_{2.5}$ standard deviation of 74.83 μg/m^3 confirm that Delhi's air quality changes dramatically across seasons rather than remaining stable throughout the year.

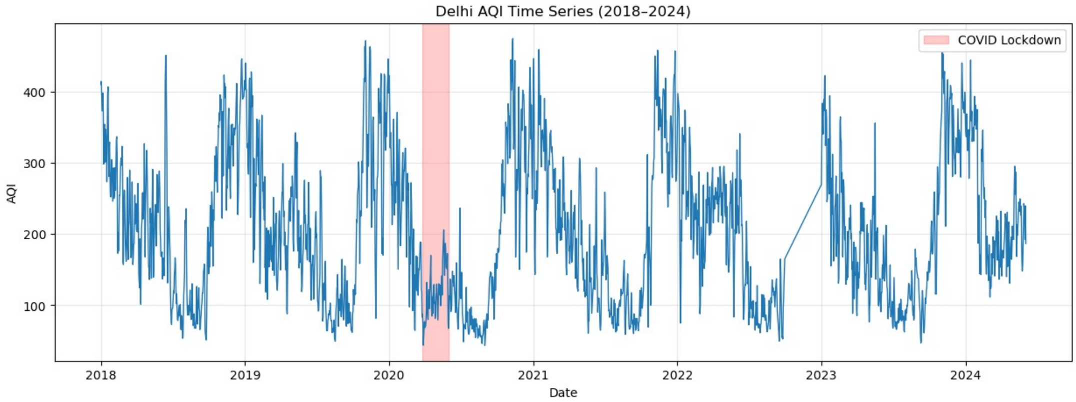


Figure 1: Timeseries graph of AQI vs time with the lockdown period highlighted

5.2 Cross-Dataset Comparison and Validation

A critical preliminary analysis compared the CPCB AQI and Embassy $PM_{2.5}$ datasets on their 2,168 shared days. The correlation between the two was exceptional: Pearson $r = 0.925$ and Spearman $r = 0.943$, both $p < 0.001$. This demonstrates that despite measuring different quantities at different locations with different instruments, both datasets track the same underlying atmospheric pollution cycle.

Seasonal means from both datasets are presented together in Table 3.

Table 3: Seasonal Mean Air Quality Values — Both Datasets

Season	**CPCB Mean AQI**	**Embassy Mean $PM_{2.5}$ (μg/m^3)**
Winter	307.26	237.9
Post-Monsoon	229.55	168.3
Pre-Monsoon	182.07	138.3
Monsoon	99.49	93.0

Both datasets show the same seasonal ranking: Winter > Post-Monsoon > Pre-Monsoon > Monsoon. The approximately 3:1 ratio between Winter and Monsoon values is consistent across both measurement approaches, independently validating the seasonal structure of Delhi's pollution cycle. Monthly mean values for both datasets are presented in Table 4. The two datasets track each other closely across all twelve months, with the Embassy station reading somewhat lower than CPCB in absolute terms — consistent with its location in the greener, less congested Chanakyapuri district.

5.3 Seasonal Structure of Air Quality

Both datasets exhibit extremely strong seasonal variability. Winter AQI values (307.26) were more than three times larger than monsoon AQI values (99.49). The Embassy data mirrors this pattern: Winter $PM_{2.5}$ (237.9 μg/m^3) was more than 2.5 times higher than Monsoon $PM_{2.5}$ (93.0 μg/m^3).

One-way ANOVA tests confirmed that seasonal differences are highly statistically significant across both datasets (Table 5).

Table 4: Monthly Mean Air Quality Values — Both Datasets

Month	**CPCB Mean AQI**	**Embassy Mean $PM_{2.5}$ ($\mu g/m^3$)**
January	313.8	252.0
February	244.4	185.2
March	185.4	147.2
April	192.0	141.3
May	186.5	140.8
June	161.4	122.2
July	100.1	94.8
August	96.5	86.8
September	101.9	97.4
October	229.5	168.3
November	344.6	256.7
December	342.4	253.1

Table 5: ANOVA Results for Seasonal Air Quality Differences

Dataset	**F-statistic**	**p-value**
CPCB AQI (4 seasons)	1137.8	< 0.001
Embassy $PM_{2.5}$ (4 seasons)	1053.3	< 0.001

These results are particularly important because emissions in Delhi do not decrease by a factor of three during monsoon months. Vehicles continue operating, industries remain active, and agricultural activity in surrounding states continues. Therefore, the magnitude of seasonal differences cannot be explained by emissions alone. Instead, atmospheric conditions appear to strongly regulate pollution accumulation. During winter, stable atmospheric conditions and weak vertical mixing trap pollutants close to the surface. During monsoon months, rainfall and stronger circulation promote atmospheric cleansing.

5.4 Rainfall and Atmospheric Cleansing

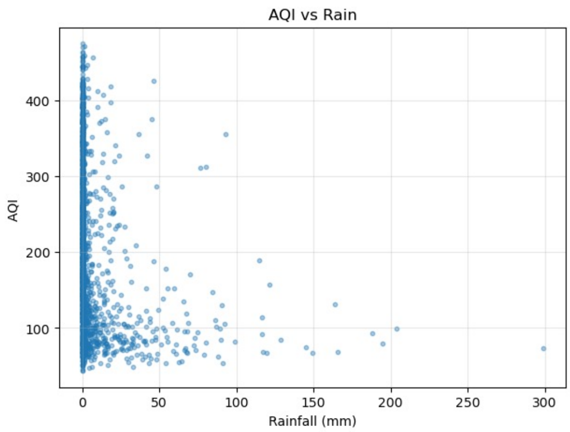
Rainfall exhibited a statistically significant negative relationship with both air quality measures. Spearman correlation was used throughout because rainfall data are zero-inflated and air quality distributions are right-skewed, violating the assumptions required for Pearson correlation. Results are presented in Table 6.

Table 6: Spearman Correlation Analysis (all $p < 0.001$)

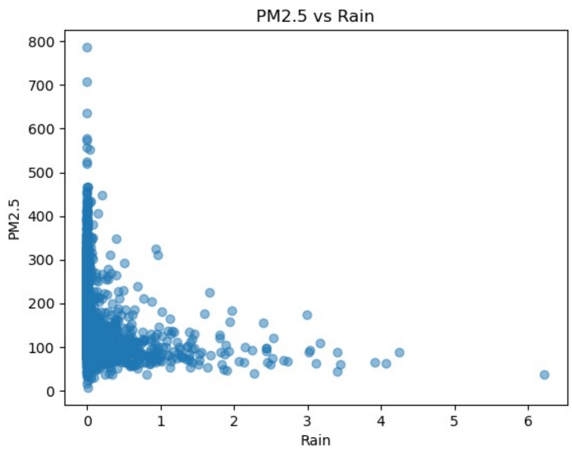
Variables	**Spearman r (AQI)**	**Spearman r (Embassy $PM_{2.5}$)**
Air quality vs Rainfall	-0.458	-0.474
Air quality vs Wind component	$+0.249$	$+0.238$

The negative rainfall correlation is consistent across both datasets, indicating that precipitation contributes to atmospheric cleansing through wet deposition. Falling raindrops remove suspended pollutants from the atmosphere, reducing both AQI and $PM_{2.5}$ concentrations.

The wind component shows a weak positive correlation with air quality in both datasets. This result reflects the fact that the wind variable in this analysis is a signed ERA5 directional component rather than wind speed magnitude. The relationship between wind and pollution is therefore directionally dependent and not interpretable as a simple dispersion effect. Further analysis using proper wind speed magnitude and direction would be needed to characterise this relationship fully.



(a) AQI vs Rainfall



(b) $PM_{2.5}$ vs Rainfall

Figure 2: Relationship between rainfall and air quality indicators in Delhi (2018–2024). The top panel shows AQI response, while the nottom panel shows $PM_{2.5}$ response.

5.5 Lagged Rainfall Effects

One of the most important findings of this study emerged when lagged rainfall effects were examined. The strongest anti-correlation between rainfall and air quality did not occur on the same day as rainfall events, but one to two days afterward (Table 7).

The consistency of this finding across both datasets is striking. CPCB AQI peaks at Lag 2 ($r = -0.568$) while Embassy $PM_{2.5}$ peaks at Lag 1 ($r = -0.534$). Both show that atmospheric cleansing effects persist for multiple days after rainfall events end. Pollutants require time to rebuild in the atmosphere after rainfall-driven removal, and wet surfaces

Table 7: Lagged Rainfall Correlations — Both Datasets (Spearman)

Lag	**AQI r**	**AQI p-value**	**Embassy r**	**Embassy p-value**
Same Day	−0.458	< 0.001	−0.474	< 0.001
Lag 1 (1 day)	−0.543	< 0.001	−0.534	< 0.001
Lag 2 (2 days)	−0.568	< 0.001	−0.521	< 0.001
Lag 3 (3 days)	−0.552	< 0.001	−0.502	< 0.001

may suppress dust resuspension in the days following precipitation.

This result is inconsistent with the null hypothesis. If pollution were controlled purely by emissions, rainfall effects would be expected to appear instantaneously and dissipate immediately. The persistent lagged structure instead suggests that the atmosphere possesses a kind of "memory," where meteorological events continue shaping air quality for days after the event itself.

Insert Figure 5 here: *Bar graph of lagged rainfall correlations for both datasets.*

5.6 COVID-19 Lockdown as a Natural Experiment

The COVID-19 national lockdown in India (25 March – 31 May 2020) provided a rare natural experiment in which anthropogenic emissions temporarily declined sharply while atmospheric processes continued operating normally. Table 8 presents lockdown comparisons for both datasets.

Table 8: Air Quality During COVID-19 Lockdown vs Normal Conditions

Condition	**CPCB Mean AQI**	**Embassy Mean $PM_{2.5}$ (μg/m^3)**
Lockdown (Mar 25 – May 31 2020)	116.65	111.21
Non-Lockdown (all other days)	207.30	162.06
Apparent Reduction	43.7%	31.4%
Season-Corrected Reduction*	40.6%	24.3%

* Season-corrected: lockdown days compared against the same calendar months (March–May) in non-lockdown years only.

Both datasets show clear, statistically significant reductions in air quality during lockdown conditions (AQI: $t = -7.13$, $p < 0.001$; Embassy $PM_{2.5}$: $t = -5.55$, $p < 0.001$). This confirms that anthropogenic emissions are critically important contributors to Delhi's air quality.

However, an important methodological note applies. The lockdown period (March–May) falls entirely within the Pre-Monsoon season, which already experiences lower pollution than winter months. Comparing lockdown days against all non-lockdown days therefore overstates the emissions reduction effect. The season-corrected comparison — lockdown days versus the same calendar months in non-lockdown years — produces a

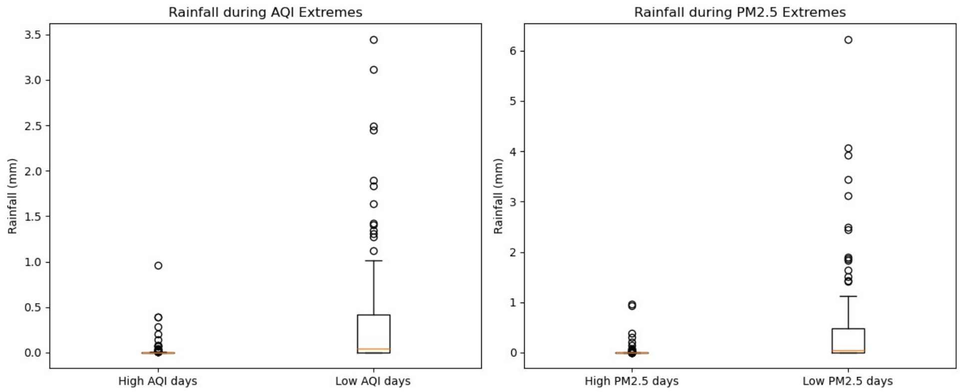


Figure 3: Boxplots comparing rainfall during extreme air pollution conditions in Delhi (2018–2024). The left panel shows rainfall distributions on days in the top 5% of AQI (high pollution) and bottom 5% of AQI (low pollution). The right panel shows the same comparison using US Embassy $PM_{2.5}$ concentrations.

more conservative estimate: 40.6% for AQI and 24.3% for Embassy $PM_{2.5}$. Even after this correction, the reductions are highly significant, confirming that emissions matter substantially. But the difference between the apparent and season-corrected reductions illustrates precisely the argument this paper makes: atmospheric conditions and the seasonal calendar independently shape air quality, separate from emissions.

5.7 Extreme Pollution Events

To further examine the atmospheric conditions associated with the worst and cleanest air quality, the cleanest and most polluted 5% of days were identified and their meteorological conditions compared (Table 9).

Table 9: Meteorological Conditions During Extreme Pollution Events

Condition	Mean Rainfall (mm)	Source
Extreme high-pollution days (top 5% AQI)	1.30	CPCB AQI
Extreme low-pollution days (bottom 5% AQI)	18.08	CPCB AQI
Extreme high-pollution days (top 5% Embassy $PM_{2.5}$)	1.50	Embassy $PM_{2.5}$
Extreme low-pollution days (bottom 5% Embassy $PM_{2.5}$)	22.25	Embassy $PM_{2.5}$

The results are consistent across both datasets. Clean days experienced dramatically more rainfall than polluted days — approximately 14 to 15 times more in both datasets. This provides strong evidence that severe pollution episodes occur not only because emissions exist, but because atmospheric conditions allow pollutants to remain trapped near the surface without removal.

High pollution days are associated with near-zero rainfall (mean ≈ 0.027 mm for AQI extremes), whereas low pollution days occur under significantly wetter conditions (mean ≈ 0.377 mm. This strong contrast (approximately an order-of-magnitude difference) supports the hypothesis that rainfall and wet deposition play a major role in reducing particulate pollution, and that meteorological conditions strongly modulate observed air quality independent of emissions.

6. Discussion

The dominant public narrative surrounding Delhi's air pollution usually assumes a relatively direct relationship between emissions and pollution severity. According to this framework, more emissions simply produce worse air quality. This study demonstrates that the relationship is considerably more complex, and that this complexity is observable consistently across two independent measurement systems covering the same city and the same time period.

The null hypothesis is rejected. Across both CPCB AQI and Embassy $PM_{2.5}$, rainfall shows statistically significant negative correlations with air quality at all four lag periods (same-day through Lag 3), with p-values far below the 0.001 threshold. Seasonal differences are so large — a 3:1 Winter-to-Monsoon ratio confirmed by both datasets — that they cannot plausibly be attributed to emissions variation alone. The ANOVA F-statistics exceed 1,000 in both datasets. The null hypothesis is rejected.

The alternate hypothesis is supported. Meteorological conditions explain a substantial and independently verifiable portion of Delhi's air quality variability. The Spearman correlation of -0.568 between AQI and Lag-2 rainfall, and -0.534 between Embassy $PM_{2.5}$ and Lag-1 rainfall, are strong effects by the standards of environmental science. These are not marginal or ambiguous findings.

The key insight of this project is not merely that weather affects pollution — atmospheric science has long recognised this. The more important finding is that atmospheric conditions actively regulate whether pollutants accumulate, persist, or are removed, and that this regulation is visible at multiple timescales simultaneously: within days (lagged rainfall effects), across seasons (monsoon cleansing), and across years (COVID lockdown as a quasi-experiment).

How does this challenge the dominant narrative? The dominant emissions-focused narrative implies a roughly linear relationship: reduce emissions, reduce pollution proportionally. The data in this study complicate this picture in at least three ways. First, the seasonal analysis shows that even without any reduction in emissions, air quality improves dramatically during the monsoon — suggesting that atmospheric conditions can dwarf the influence of emissions at certain times of year. Second, the lagged rainfall results demonstrate that meteorological events have consequences that outlast the event itself, reshaping the pollution environment for multiple days. Third, even during the COVID lockdown — the closest thing to a controlled emissions experiment available — the season-corrected reduction was only around 24–41%, meaning that more than half of Delhi's "typical" pollution level persisted even when emissions were substantially curtailed. The atmospheric accumulation component remained.

The cross-dataset comparison adds an important dimension. The Embassy data reads systematically lower than CPCB in absolute terms — as expected for a station in a greener, less traffic-dense area. However, the patterns are nearly identical: the same seasonal structure, the same lagged rainfall effects, the same lockdown response. This convergence across an independent instrument, independent location, and independent measurement methodology substantially strengthens the credibility of the findings. It is very unlikely that both datasets would independently show the same atmospheric structure if the underlying relationships were artefacts of data collection.

Together, these findings suggest that Delhi's air pollution should be understood as the product of interactions between emissions and atmospheric physics rather than emissions alone:

$$\text{Observed Air Quality} = f(\text{Emissions, Meteorology}) \tag{1}$$

Pollution severity emerges from this interaction. An emissions crisis and an atmospheric accumulation crisis coexist.

7. Conclusion

Using CPCB AQI observations and US Embassy $PM_{2.5}$ data together with ERA5 meteorological data between 2018 and 2024, this study investigated how atmospheric conditions influence Delhi's air quality. Two independent datasets, measuring different quantities at different locations with different instruments, consistently produced the same findings:

- The null hypothesis is rejected. Rainfall and seasonal atmospheric conditions have a highly significant negative relationship with Delhi's air quality across both datasets.
- Strong seasonal AQI and $PM_{2.5}$ variability exists, with Winter values more than three times higher than Monsoon values in both datasets.
- Rainfall is associated with reductions in air quality, consistent with wet deposition processes that remove particulate matter and soluble gases from the atmosphere. However, this study remains observational, and therefore these relationships are interpreted as statistical associations rather than direct causal estimates with effects that persist for two to three days after storm events — stronger than same-day rainfall effects — consistent across both measurement systems.
- Anthropogenic emissions matter: COVID lockdown conditions produced significant reductions in both AQI (40.6%) and $PM_{2.5}$ (24.3%), even after controlling for seasonal confounding.
- Extreme clean days are associated with 14–15 times more rainfall than extreme polluted days, in both datasets independently.
- The two datasets correlate at $r = 0.925$ (Pearson) and $r = 0.943$ (Spearman), confirming they track the same atmospheric signal.

These findings challenge the dominant narrative that explains Delhi's pollution primarily through emissions alone. Delhi's pollution crisis is therefore not only an emissions crisis, but also an atmospheric accumulation crisis — a conclusion supported independently by two different measurement approaches covering the same city and the same years.

References

References

[1] Central Pollution Control Board (CPCB). *National Air Quality Monitoring Programme*. Available at: https://cpcb.nic.in/

[2] Hersbach, H. et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146, 1999–2049.

[3] Mahato, S., Pal, S., and Ghosh, K. (2020). Effect of lockdown amid COVID-19 pandemic on air quality of the megacity Delhi. *Science of the Total Environment*, 730, 139086.

[4] Tai, A. P. K., Mickley, L. J., and Jacob, D. J. (2010). Correlations between fine particulate matter and meteorological variables. *Atmospheric Environment*, 44(32), 3976–3984.

[5] US Embassy New Delhi. *Air Quality Monitoring Data*. Available at: https://in.usembassy.gov/embassy-consulates/new-delhi/air-quality-data/

[6] World Health Organization (WHO). *Global Air Quality Guidelines*. Available at: https://www.who.int/

[7] Guttikunda, S. K. and Gurjar, B. R. (2014). Role of meteorology in seasonality of air pollution in megacity Delhi, India. *Environmental Monitoring and Assessment*, 184, 3199–3211.

[8] Seinfeld, J. H. and Pandis, S. N. (2016). *Atmospheric Chemistry and Physics: From Air Pollution to Climate Change*. 3rd ed., Wiley.

[9] Jacob, D. J. (1999). *Introduction to Atmospheric Chemistry*. Princeton University Press.

[10] Ghosh, A. et al. (2021). Meteorological factors and urban air quality during the COVID-19 lockdown in Delhi. *Aerosol and Air Quality Research*, 21, 200397.

[11] Zhang, L., Gong, S., Padro, J., and Barrie, L. (2001). A size-segregated particle dry deposition scheme for an atmospheric aerosol module. *Atmospheric Environment*, 35(3), 549–560.